Arm64 F4/F5 defaults — run report

Run `run-20261022T0705Z` · workflow `arm64-f4-defaults.yml`, GitHub Actions run 39732230961, attempt 1 · PR merge commit `d6b94c91f` (branch head `c034f1639`) · 2026-10-22

What ran

One binary per host and tool, arms by environment only, arm order rotated inside every round (`scripts/run_inh.sh`, `scripts/run_gf2.sh`). Decision rule: `scripts/DECISION_RULE.md`, committed before any Arm data.

- **Inherited F4 / F6-IC** (`f6_ic_probe --engines f4,f6p --reps 3`), nine cells as in `research/inherited_f4_eager_runs_20261016`, 5 rounds. Arms: **on** (defaults: eager rewrite through shift runs, gate-first F6-IC order), **noeager** (`KIC_F4_EAGER=0`: lazy rewrite with the bit scatter, what Arm hosts ran before #2043), **nogate** (`KIC_F6_GATE_FIRST=0`). CPU = probe `cpu_ms` of the arm total.
- **GF(2)** (`f4_f2_bench`), 3 rounds: F5 cases ($n \leq 24$, Reduced output) at 1 thread and at all cores; F4 cases ($n \leq 20$) at 1 thread. Arms: **t4** (default, 4 Gray-code tables), **t2**, **t3** (`KIC_GF2_TABLES`), **reuse** (`KIC_GF2_REUSE_TABLE=1`). Time = bench `wall_ms`.

Hosts (`*/host.txt`):

- `ubuntu-24.04-arm`: Arm Neoverse-N2, 4 vCPU, L2 1 MiB per core, L3 128 MiB, Linux 6.17 (Azure), rustc 1.99.1; load 5.1 at start, 1.0–1.2 later. Shared, unpinned (L0).
- `macos-latest`: Apple M1 (virtual), 3 vCPU, L2 12 MiB, macOS kernel 25.6, rustc 1.99.1; the VM reported load average 36–50 at the start of the inherited arms and 1.8–14 at the end of the GF(2) arms. Shared, unpinned (L0); its timings carry much wider round-to-round spread.

Runs: inherited 135 per host, GF(2) 36 per host; every run exited 0 on both hosts.

Exactness

On both hosts all 80 targets have the same decision tuple in every arm and round; each arm's word count is identical across rounds; the GF(2) fingerprints (basis / rows / row space, terms, rank, word XORs) agree across all arms and rounds for all 21 case keys. The word counts are identical on the two hosts, and **on** / **noeager** charge exactly what the x86 run `research/inherited_f4_eager_runs_20261016` recorded for **runs** = **bmi2** / **scatter**.

Inherited F4 / F6-IC

Ratios are medians over 5 rounds (arm ÷ default); the bracket is the worst–best round pairing. Units: per-arm total `word_xors`.

`ubuntu-24.04-arm`

cell	engine	CPU on ms	noeager/on	nogate/on	units on	units noeager	units nogate
K1_17	f4	455.1	1.12 (1.11–1.12)	1.00 (0.99–1.00)	18019804	15109049	18019804
K1_17	f6p	165.2	1.11 (1.10–1.12)	1.85 (1.83–1.87)	10830979	8865230	15514870
K1_23	f4	867.8	1.28 (1.27–1.28)	1.00 (0.99–1.00)	62213299	52572523	62213299
K1_23	f6p	357.6	1.21 (1.20–1.22)	2.10 (2.09–2.11)	38122485	32069197	59881009
K0_23	f4	2516.8	1.30 (1.29–1.31)	1.00 (0.99–1.01)	167017420	148425559	167017420
K0_23	f6p	819.3	1.36 (1.35–1.37)	2.01 (1.99–2.02)	81022008	67544699	132377695
K0_31	f4	2593.1	1.61 (1.61–1.63)	1.00 (0.99–1.01)	290412310	264887288	290412310
K0_31	f6p	1188.7	1.57 (1.56–1.58)	2.01 (2.00–2.03)	164736149	146257576	285541596
K0_19	f4	334.5	1.13 (1.13–1.14)	1.00 (1.00–1.00)	17250812	15282173	17250812
K0_19	f6p	119.6	1.12 (1.11–1.13)	1.69 (1.68–1.70)	10226773	8837211	13660081
K1_19	f4	581.2	1.20 (1.20–1.21)	1.00 (0.99–1.01)	30159678	25782851	30159678
K1_19	f6p	236.2	1.15 (1.14–1.15)	1.92 (1.91–1.93)	19117122	16026687	28309592
K1_17d7	f4	528.0	1.08 (1.08–1.09)	1.00 (1.00–1.00)	22448248	19473310	22448248
K1_17d7	f6p	157.5	1.09 (1.08–1.10)	1.82 (1.81–1.83)	10867029	8987079	15297452
K1_29	f4	1862.8	1.52 (1.51–1.52)	1.00 (1.00–1.00)	185919270	165390163	185919270
K1_29	f6p	870.0	1.45 (1.45–1.46)	1.96 (1.95–1.97)	112675706	98192767	182738266
K0_37	f4	4579.1	1.69 (1.68–1.71)	1.00 (1.00–1.00)	652414757	604262254	652414757
K0_37	f6p	2138.3	1.62 (1.61–1.65)	2.02 (2.01–2.03)	376487242	342788148	644803622

`macos-latest`

cell	engine	CPU on ms	noeager/on	nogate/on	units on	units noeager	units nogate
K1_17	f4	604.8	1.18 (0.81–1.76)	1.04 (0.71–1.53)	18019804	15109049	18019804
K1_17	f6p	222.6	1.11 (0.94–1.59)	1.71 (1.41–2.54)	10830979	8865230	15514870
K1_23	f4	1161.5	1.33 (1.20–1.45)	0.98 (0.82–1.06)	62213299	52572523	62213299
K1_23	f6p	492.4	1.27 (1.02–1.53)	2.05 (1.55–2.30)	38122485	32069197	59881009
K0_23	f4	3348.0	1.33 (1.03–1.57)	1.01 (0.84–1.10)	167017420	148425559	167017420

K0_23	f6p	1020.6	1.61 (1.06–2.00)	2.15 (1.57–2.67)	81022008	67544699	132377695
K0_31	f4	3841.7	1.42 (1.20–2.18)	0.93 (0.82–1.32)	290412310	264887288	290412310
K0_31	f6p	1751.9	1.53 (1.26–2.01)	1.90 (1.74–2.66)	164736149	146257576	285541596
K0_19	f4	502.7	1.17 (0.83–1.40)	0.98 (0.74–1.20)	17250812	15282173	17250812
K0_19	f6p	173.2	1.19 (0.84–1.67)	1.76 (1.52–2.31)	10226773	8837211	13660081
K1_19	f4	883.8	1.15 (0.87–1.66)	1.06 (0.70–1.21)	30159678	25782851	30159678
K1_19	f6p	401.5	0.97 (0.71–1.54)	1.75 (1.21–2.27)	19117122	16026687	28309592
K1_17d7	f4	821.4	1.09 (0.86–1.44)	0.99 (0.78–1.28)	22448248	19473310	22448248
K1_17d7	f6p	246.0	1.07 (0.79–1.43)	1.62 (1.44–2.56)	10867029	8987079	15297452
K1_29	f4	2565.2	1.44 (1.18–2.02)	0.96 (0.79–1.32)	185919270	165390163	185919270
K1_29	f6p	1091.7	1.52 (1.13–2.00)	2.13 (1.53–2.67)	112675706	98192767	182738266
K0_37	f4	5871.8	1.87 (1.47–2.13)	1.01 (0.83–1.27)	652414757	604262254	652414757
K0_37	f6p	2648.5	1.78 (1.57–2.30)	2.10 (1.63–2.57)	376487242	342788148	644803622

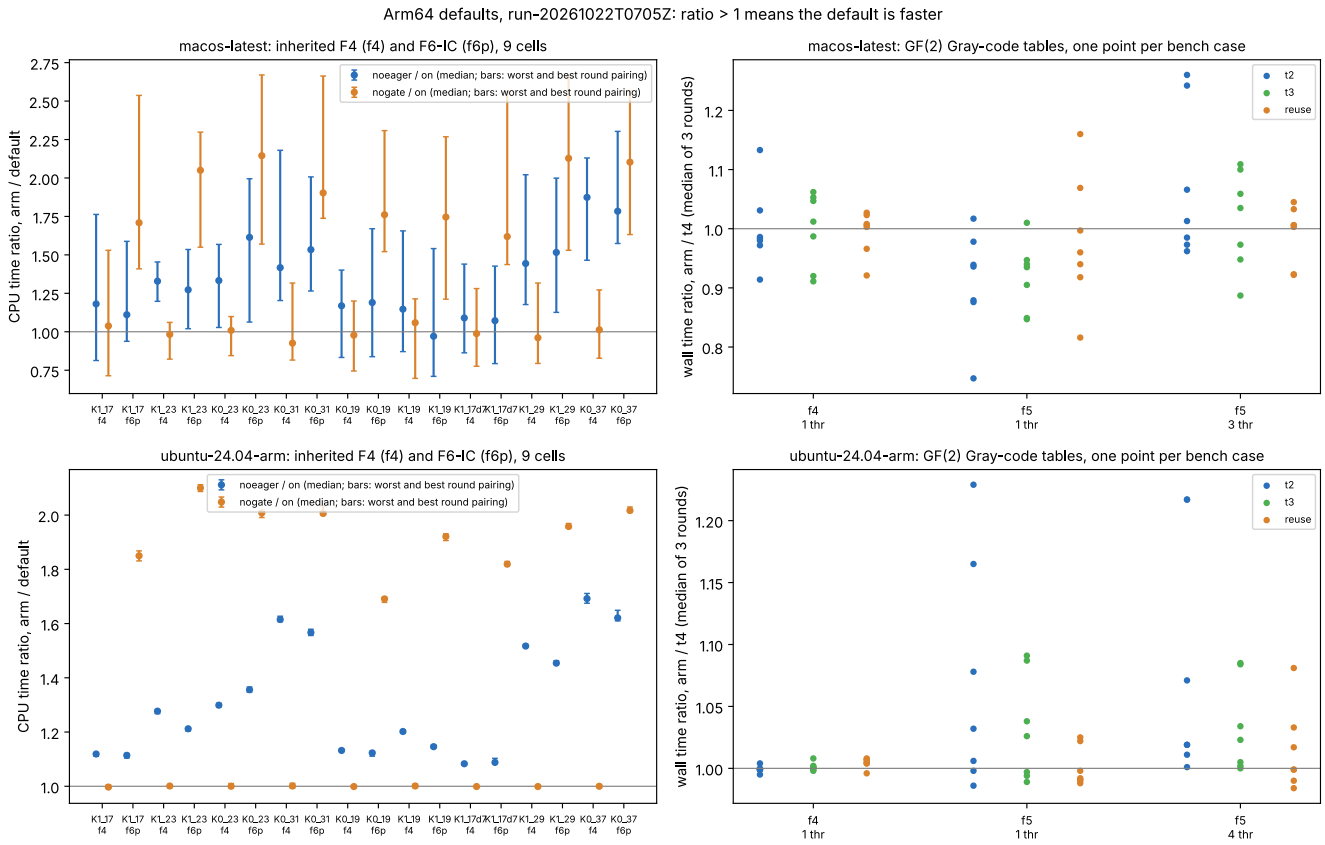


Figure 1: Left: CPU ratio of each switched-off arm over the default per cell and engine (5 rounds; bars: worst and best round pairing). Right: GF(2) wall ratio arm / t4 per bench case (median of 3 rounds). run-20261022T0705Z.

GF(2) tables

Geometric mean, minimum and maximum over the cases of the per-case median ratio arm ÷ **t4** (below 1: the arm is faster).

ubuntu-24.04-arm

tool	threads	arm	geomean arm/t4	min	max	cases
f4	1	reuse	1.004	0.996	1.008	7
f4	1	t2	0.999	0.995	1.004	7
f4	1	t3	1.001	0.998	1.008	7
f5	1	reuse	1.001	0.988	1.025	7
f5	1	t2	1.067	0.986	1.229	7
f5	1	t3	1.031	0.989	1.091	7
f5	4	reuse	1.014	0.984	1.081	7

f5	4	t2	1.076	1.001	1.217	7
f5	4	t3	1.033	1.000	1.085	7

`macos-latest`

tool	threads	arm	geomean arm/t4	min	max	cases
f4	1	reuse	0.993	0.921	1.027	7
f4	1	t2	0.998	0.914	1.133	7
f4	1	t3	0.997	0.911	1.062	7
f5	1	reuse	0.975	0.816	1.160	7
f5	1	t2	0.906	0.747	1.017	7
f5	1	t3	0.917	0.847	1.010	7
f5	3	reuse	0.990	0.922	1.045	7
f5	3	t2	1.065	0.962	1.260	7
f5	3	t3	1.013	0.887	1.109	7

The pre-registered rule applied

1. **Eager rewrite through shift runs** (keep if `noeager/on` ≥ 1 on ≥ 6 of 9 cells per engine per host): Linux 9/9 (f4) and 9/9 (f6p), 1.08–1.69×; macOS 9/9 (f4) and 8/9 (f6p), 0.97–1.87×. **Stays the default.**
2. **Gate-first F6-IC** (keep if `nogate/on` ≥ 1 on ≥ 7 of 9 cells for f6p per host): Linux 9/9, 1.69–2.10×; macOS 9/9, 1.62–2.15×. **Stays the default.** The f4 engine (no node oracle) reads 1.00 on every Linux cell and 0.93–1.06 on macOS, the A/A spread of each host.
3. **GF(2) tables / reuse** (an Arm default only if one arm is ≥ 5 % faster in the geometric mean at 1 thread and at all cores on both hosts, no case below 0.97×): Linux — no arm is faster than **t4** in any group (geomeans 0.999–1.076). macOS — **t2** and **t3** are 1.10× and 1.09× faster at 1 thread on F5, but 0.94× and 0.99× at 3 threads. No arm meets the rule; **the x86-tuned defaults stand.**

Observations and limits

- These are CPU or wall times of shared, unpinned CI virtual machines; the macOS VM ran under heavy reported load during the inherited arms, so its brackets span 1 on many cells even where the median is far from it. The Linux runner's brackets lie within about ±2 % of the median.
- One run per host; no second attempt and no isolated host. The ratios describe these runners, not Arm hardware in general.
- Nothing here is an index-calculus total or a comparison against rho; it is a check that two solver defaults are right on another architecture, and that one constant needs no Arm-specific value.

Raw records: `ubuntu-24.04-arm/{inh,gf2}/raw/`, `macos-latest/{inh,gf2}/raw/`; tables `*/inh.tsv`, `*/gf2.tsv`; `summary.txt` (from `summarize.py`); figure from `plot.py`.